Experiment No:1

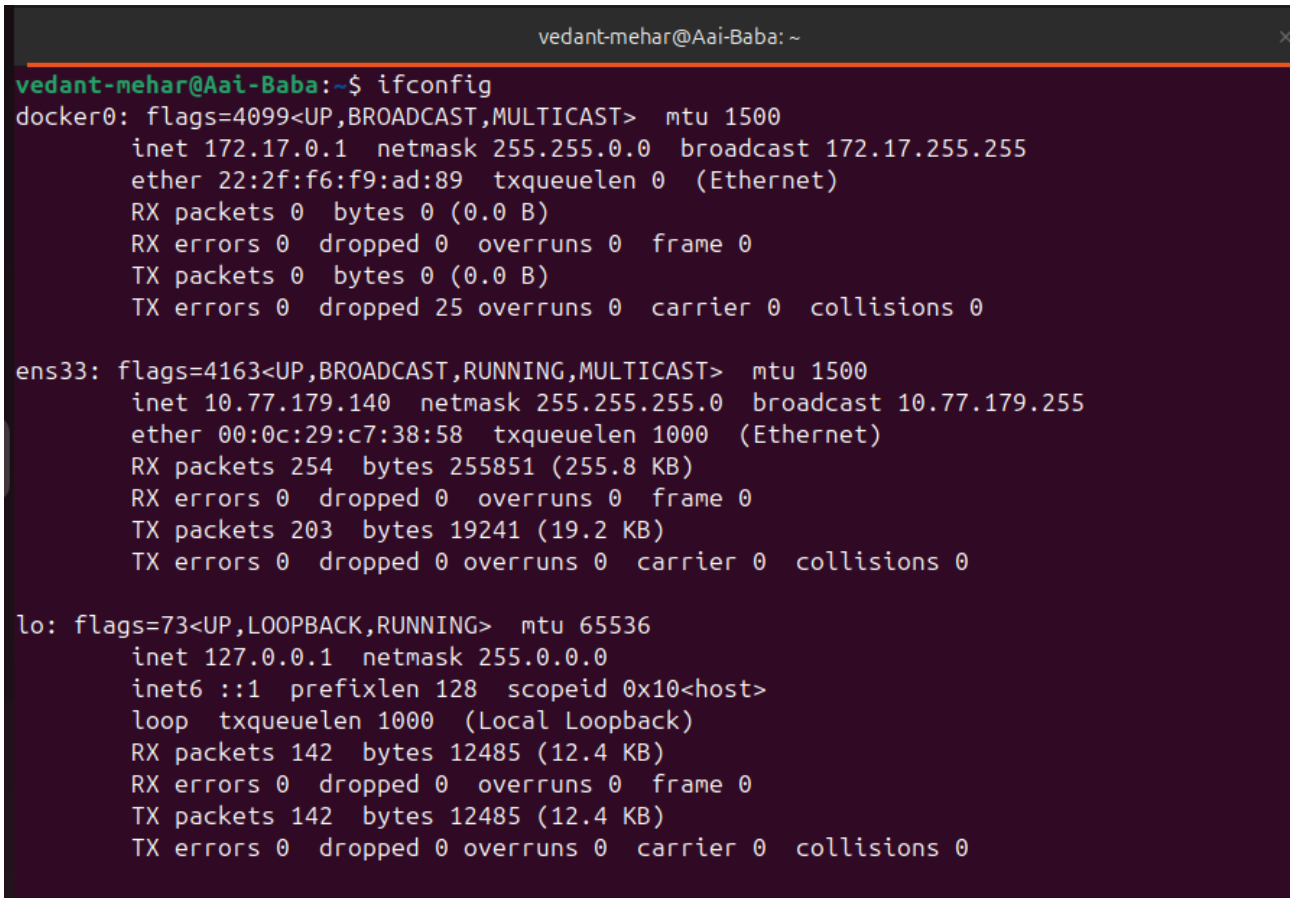
Title: Study Networking commands

Problem Statement: Study and implement various networking commands on terminal

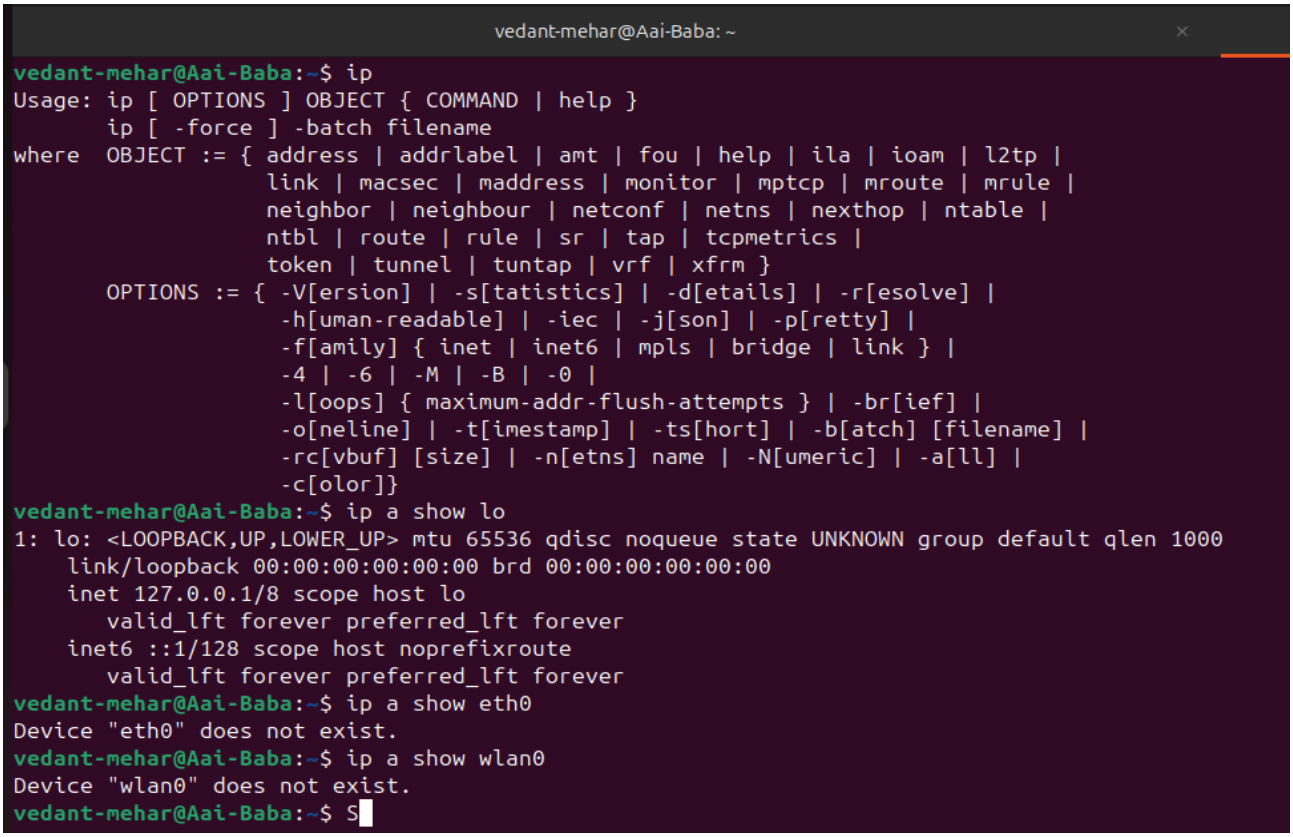
Description:
Linux networking commands are used extensively to inspect, analyze, maintain, and troubleshoot the network/s connected to the system.

Name: Vedant Narayan Mehar Roll No.: 13 PRN: 12413063

1. Ifconfig (Interface Configurator)



2. ip



3. traceroute

```
Preparing to unpack .../inetutils-traceroute_2%3a2.5-3ubuntu4.2_amd64.deb ...
Unpacking inetutils-traceroute (2:2.5-3ubuntu4.2) ...
Setting up inetutils-telnet (2:2.5-3ubuntu4.2) ...
Setting up inetutils-traceroute (2:2.5-3ubuntu4.2) ...
update-alternatives: using /usr/bin/inetutils-traceroute to provide /usr/bin/traceroute (traceroute) in auto mode
Setting up telnet (0.17+2.5-3ubuntu4.2) ...
Processing triggers for man-db (2.12.0-4build2) ...
vedant-mehar@Aai-Baba:~$ traceroute google.com
traceroute to google.com (192.178.211.100), 64 hops max
  1   10.77.179.135  5.152ms  1.918ms  2.500ms
  2   *  *  *
  3   255.0.0.2  30.577ms  19.633ms  *
  4   255.0.0.3  27.289ms  18.955ms  18.861ms
  5   192.168.227.67  17.819ms  20.968ms  31.120ms
  6   192.168.92.164  17.902ms  192.168.92.166  28.352ms  21.036ms
  7   *  *  *
  8   *  *  *
  9   * ^C
vedant-mehar@Aai-Baba:~$ traceroute google.com
traceroute to google.com (192.178.211.138), 64 hops max
  1   10.77.179.135  29.580ms  3.950ms  2.910ms
  2   *  *  *
  3   255.0.0.2  34.441ms  19.805ms  17.498ms
  4   255.0.0.3  23.173ms  17.463ms  16.765ms
  5   192.168.227.66  19.735ms  20.025ms  17.143ms
  6   192.168.92.164  21.303ms  18.782ms  26.005ms
  7   *  *  *
  8   *  *  *
  9   142.250.169.146  44.717ms  26.047ms  21.339ms
 10   *  *  *
 11   172.253.177.90  28.348ms  31.159ms  15.494ms
 12   192.178.254.31  28.625ms  20.823ms  30.407ms
 13   *  *  *
 14   192.178.254.67  26.996ms  16.652ms  20.627ms
 15   192.178.254.121  17.797ms  20.035ms  20.609ms
 16   *  *  *
 17   *
 * ^C
vedant-mehar@Aai-Baba:~$
```

4. tracepath

```
vedant-mehar@Aai-Baba:~$ tracepath mindmajix.com
 1?: [LOCALHOST]                      pmtu 1500
 1:  _gateway                                              4.382ms
 1:  _gateway                                              4.025ms
 2:  _gateway                                              5.125ms pmtu 1372
 2:  no reply
 3:  no reply
 4:  255.0.0.3                                            40.103ms asymm  5
 5:  192.168.227.67                                       25.020ms asymm  6
 6:  192.168.92.162                                       24.232ms asymm  7
 7:  no reply
 8:  no reply
 9:  no reply
10:  no reply
11:  no reply
12:  no reply
13:  no reply
14:  no reply
15:  no reply
16:  no reply
17:  no reply
18:  no reply
```

5. ping

```
vedant-mehar@Aai-Baba:~$ ping
ping: usage error: Destination address required
vedant-mehar@Aai-Baba:~$ ping google.com
PING google.com (192.178.211.113) 56(84) bytes of data.
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=1 ttl=105 time=21.0 ms
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=2 ttl=105 time=29.1 ms
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=3 ttl=105 time=28.4 ms
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=4 ttl=105 time=45.2 ms
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=5 ttl=105 time=28.0 ms
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=6 ttl=105 time=30.4 ms
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=7 ttl=105 time=21.7 ms
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=8 ttl=105 time=23.9 ms
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=9 ttl=105 time=31.8 ms
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=10 ttl=105 time=28.2 ms
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=11 ttl=105 time=31.5 ms
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=12 ttl=105 time=27.0 ms
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=13 ttl=105 time=24.4 ms
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=14 ttl=105 time=18.2 ms
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=15 ttl=105 time=19.2 ms
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=16 ttl=105 time=30.3 ms
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=17 ttl=105 time=29.3 ms
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=18 ttl=105 time=29.7 ms
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=19 ttl=105 time=33.0 ms
64 bytes from lcbomp-in-f113.1e100.net (192.178.211.113): icmp_seq=20 ttl=105 time=46.9 ms
^C
--- google.com ping statistics ---
23 packets transmitted, 20 received, 13.0435% packet loss, time 22057ms
rtt min/avg/max/mdev = 18.218/28.864/46.896/7.070 ms
vedant-mehar@Aai-Baba:~$
```

6. netstat

```
vedant-mehar@Aai-Baba:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 Aai-Baba:43574          151.101.209.91:https    ESTABLISHED
tcp        0      0 Aai-Baba:59068          151.101.1.91:https      ESTABLISHED
tcp        0      0 Aai-Baba:36732          93.243.107.34.bc.:https ESTABLISHED
tcp        0      0 Aai-Baba:32938          123.208.120.34.bc:https ESTABLISHED
tcp        0      0 Aai-Baba:43582          151.101.209.91:https    ESTABLISHED
tcp        0      0 Aai-Baba:43580          151.101.209.91:https    ESTABLISHED
udp        0      0 Aai-Baba:bootpc         _gateway:bootps         ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     30149    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     29958    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     23358
unix  3      [ ]         STREAM     CONNECTED     32515
unix  3      [ ]         STREAM     CONNECTED     33563    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     20491    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     32307    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     31895
unix  3      [ ]         STREAM     CONNECTED     32867    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     32799
unix  3      [ ]         STREAM     CONNECTED     31574    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     32528    /run/user/1000/wayland-0
unix  3      [ ]         STREAM     CONNECTED     30140    /run/user/1000/bus
unix  2      [ ]         DGRAM      CONNECTED     20784
unix  3      [ ]         STREAM     CONNECTED     32534    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     30933
unix  3      [ ]         STREAM     CONNECTED     32209    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     33612    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     32877
unix  3      [ ]         STREAM     CONNECTED     32800    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     24306    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     32103
unix  3      [ ]         STREAM     CONNECTED     30563    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     30931
unix  3      [ ]         STREAM     CONNECTED     32535
unix  3      [ ]         STREAM     CONNECTED     29989    /run/systemd/journal/stdout
unix  2      [ ]         DGRAM      CONNECTED     29802
```

```
    TCPLossProbes: 19
    TCPBacklogCoalesce: 147
    TCPDSACKOldSent: 35
    TCPDSACKOfoSent: 1
    TCPDSACKRecv: 16
    20 connections reset due to unexpected data
    7 connections reset due to early user close
    TCPDSACKIgnoredNoUndo: 7
    TCPSackMerged: 1
    TCPSackShiftFallback: 134
    IPReversePathFilter: 3
    TCPRcvCoalesce: 8306
    TCPOFOQueue: 138
    TCPOFOMerge: 1
    TCPAutoCorking: 17
    TCPSynRetrans: 1
    TCPOrigDataSent: 2055
    TCPHystartDelayDetect: 3
    TCPHystartDelayCwnd: 61
    TCPDelivered: 2145
    TCPAckCompressed: 11
    TcpTimeoutRehash: 1
    TCPDSACKRecvSegs: 16
IpExt:
    InMcastPkts: 268
    OutMcastPkts: 85
    InOctets: 52935784
    OutOctets: 1870572
    InMcastOctets: 34076
    OutMcastOctets: 8274
    InNoECTPkts: 45847
MPTcpExt:
vedant-mehar@Aai-Baba:~$ netstat -r
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
default         _gateway        0.0.0.0         UG        0 0          0 ens33
10.77.179.0     0.0.0.0         255.255.255.0   U         0 0          0 ens33
172.17.0.0      0.0.0.0         255.255.0.0     U         0 0          0 docker0
vedant-mehar@Aai-Baba:~$
```

7. ss

```
vedant-mehar@Aai-Baba:~$ ss
Netid       State       Recv-Q       Send-Q                                  Local Address:Port                  Peer Address:Port       Process
u_str       ESTAB       0            0                      /run/systemd/journal/stdout 30149                      * 31008
u_str       ESTAB       0            0                      /run/systemd/journal/stdout 29958                      * 30804
u_str       ESTAB       0            0                                          * 23358                            * 23359
u_str       ESTAB       0            0                                          * 32515                            * 33914
u_str       ESTAB       0            0                             /run/user/1000/bus 33563                        * 33562
u_str       ESTAB       0            0                      /run/systemd/journal/stdout 20491                      * 20490
u_str       ESTAB       0            0                             /run/user/1000/bus 32307                        * 32306
u_str       ESTAB       0            0                                          * 31895                            * 32908
u_str       ESTAB       0            0                      /run/systemd/journal/stdout 32867                      * 32866
u_str       ESTAB       0            0                                          * 32799                            * 32800
u_str       ESTAB       0            0                      /run/systemd/journal/stdout 31574                      * 30487
u_str       ESTAB       0            0                        /run/user/1000/wayland-0 32528                       * 32527
u_str       ESTAB       0            0                             /run/user/1000/bus 30140                        * 30991
u_dgr       ESTAB       0            0                                          * 20784                            * 11642
u_str       ESTAB       0            768                           /run/user/1000/bus 32534                        * 32533
u_str       ESTAB       0            0                                          * 30933                            * 30937
u_str       ESTAB       0            0                      /run/systemd/journal/stdout 32209                      * 33539
u_str       ESTAB       0            0                             @/tmp/.X11-unix/X0 33612                        * 31931
u_str       ESTAB       0            0                                          * 32877                            * 31859
u_str       ESTAB       0            0                      /run/systemd/journal/stdout 32800                      * 32799
u_str       ESTAB       0            0                      /run/dbus/system_bus_socket 24306                      * 24305
u_str       ESTAB       0            0                                          * 32103                            * 33347
u_str       ESTAB       0            0                             /run/user/1000/bus 30563                        * 30562
u_str       ESTAB       0            0                                          * 30931                            * 30943
u_str       ESTAB       0            0                                          * 32535                            * 32536
u_str       ESTAB       0            0                      /run/systemd/journal/stdout 29989                      * 30868
u_dgr       ESTAB       0            0                                          * 29802                            * 11640
u_str       ESTAB       0            0                          /run/user/1000/at-spi/bus 33561                    * 33560
u_str       ESTAB       0            0                             /run/user/1000/bus 32122                        * 32121
u_str       ESTAB       0            0                                          * 30486                            * 31573
u_str       ESTAB       0            0                      /run/dbus/system_bus_socket 24160                      * 24159
u_str       ESTAB       0            0                                          * 31915                            * 32942
u_str       ESTAB       0            0                                          * 32824                            * 32825
u_str       ESTAB       0            0                             /run/user/1000/bus 30123                        * 30978
u_str       ESTAB       0            0                      /run/systemd/journal/stdout 29980                      * 30824
u_str       ESTAB       0            0                                          * 32470                            * 33857
u_str       ESTAB       0            0                                          * 32956                            * 31920
u_str       ESTAB       0            0                      /run/systemd/journal/stdout 32890                      * 31867
```

8. dig

```
vedant-mehar@Aai-Baba:~$ dig google.com

; <<>> DiG 9.18.30-0ubuntu0.24.04.2-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 2933
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             150     IN      A       142.250.134.138
google.com.             150     IN      A       142.250.134.100
google.com.             150     IN      A       142.250.134.139
google.com.             150     IN      A       142.250.134.101
google.com.             150     IN      A       142.250.134.113
google.com.             150     IN      A       142.250.134.102

;; Query time: 37 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Tue Jul 07 13:16:53 IST 2026
;; MSG SIZE  rcvd: 135

vedant-mehar@Aai-Baba:~$
```

9. nslookup

```
vedant-mehar@Aa...  ×   vedant-mehar@Aa...  ×   vedant-meha

vedant-mehar@Aai-Baba:~$ nslookup mindmajix.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   mindmajix.com
Address: 216.150.1.1
Name:   mindmajix.com
Address: 216.150.1.193
Name:   mindmajix.com
Address: 64:ff9b::d896:1c1
Name:   mindmajix.com
Address: 64:ff9b::d896:1081

vedant-mehar@Aai-Baba:~$
```

10. route

```
vedant-mehar@Aai-Baba:~$ route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
default         _gateway        0.0.0.0         UG    100    0        0 ens33
10.77.179.0     0.0.0.0         255.255.255.0   U     100    0        0 ens33
172.17.0.0      0.0.0.0         255.255.0.0     U     0      0        0 docker0
vedant-mehar@Aai-Baba:~$
```

11. host

```
vedant-mehar@Aai-Baba:~$ host mindmajix.com
mindmajix.com has address 216.150.1.193
mindmajix.com has address 216.150.1.1
mindmajix.com has IPv6 address 64:ff9b::d896:1041
mindmajix.com has IPv6 address 64:ff9b::d896:101
mindmajix.com mail is handled by 10 alt4.aspmx.l.google.com.
mindmajix.com mail is handled by 5 alt1.aspmx.l.google.com.
mindmajix.com mail is handled by 5 alt2.aspmx.l.google.com.
mindmajix.com mail is handled by 1 aspmx.l.google.com.
mindmajix.com mail is handled by 10 alt3.aspmx.l.google.com.
vedant-mehar@Aai-Baba:~$
```

12. arp

```
vedant-mehar@Aai-Baba:~$ arp
Address                  HWtype  HWaddress           Flags Mask            Iface
_gateway                 ether   7a:d3:64:b4:f0:22   C                     ens33
vedant-mehar@Aai-Baba:~$
```

13. iwconfig

```
vedant-mehar@Aai-Baba:~$ iwconfig
lo        no wireless extensions.

ens33     no wireless extensions.

docker0   no wireless extensions.

vedant-mehar@Aai-Baba:~$
```

14. hostname

```
vedant-mehar@Aai-Baba:~$ hostname
Aai-Baba
vedant-mehar@Aai-Baba:~$
```

15. curl or wget

```
vedant-mehar@Aai-Baba:~$  curl -O google.com/doodles/childrens-day-2014-multiple-countr
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100   265  100   265    0     0   1917      0 --:--:-- --:--:-- --:--:--  1934
vedant-mehar@Aai-Baba:~$
```

```
vedant-mehar@Aai-Baba:~$ wget google.com/doodles/new-years-day-2012
--2026-07-07 13:30:00--  http://google.com/doodles/new-years-day-2012
Resolving google.com (google.com)... 142.250.134.113, 142.250.134.101, 142.250.134.138, ...
Connecting to google.com (google.com)|142.250.134.113|:80... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://www.google.com/doodles/new-years-day-2012 [following]
--2026-07-07 13:30:00--  https://www.google.com/doodles/new-years-day-2012
Resolving www.google.com (www.google.com)... 142.251.157.119, 142.251.155.119, 142.251.150.119, ...
Connecting to www.google.com (www.google.com)|142.251.157.119|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://doodles.google/doodle/new-years-day-2012/ [following]
--2026-07-07 13:30:01--  https://doodles.google/doodle/new-years-day-2012/
Resolving doodles.google (doodles.google)... 142.251.106.141, 2404:6800:4013:800::8d
Connecting to doodles.google (doodles.google)|142.251.106.141|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 95847 (94K) [text/html]
Saving to: ‘new-years-day-2012’

new-years-day-2012                  100%[=============================================================================>]  93.60K  --.-KB/s    in 0.07s

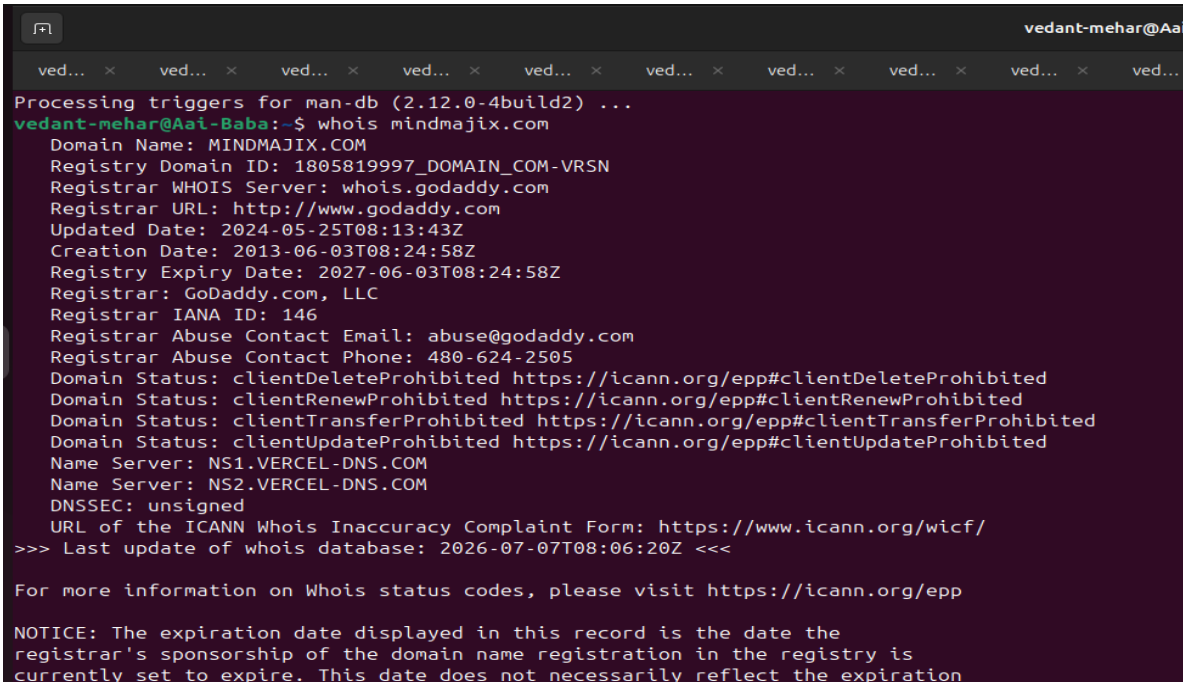
2026-07-07 13:30:02 (1.23 MB/s) - ‘new-years-day-2012’ saved [95847/95847]

vedant-mehar@Aai-Baba:~$
```

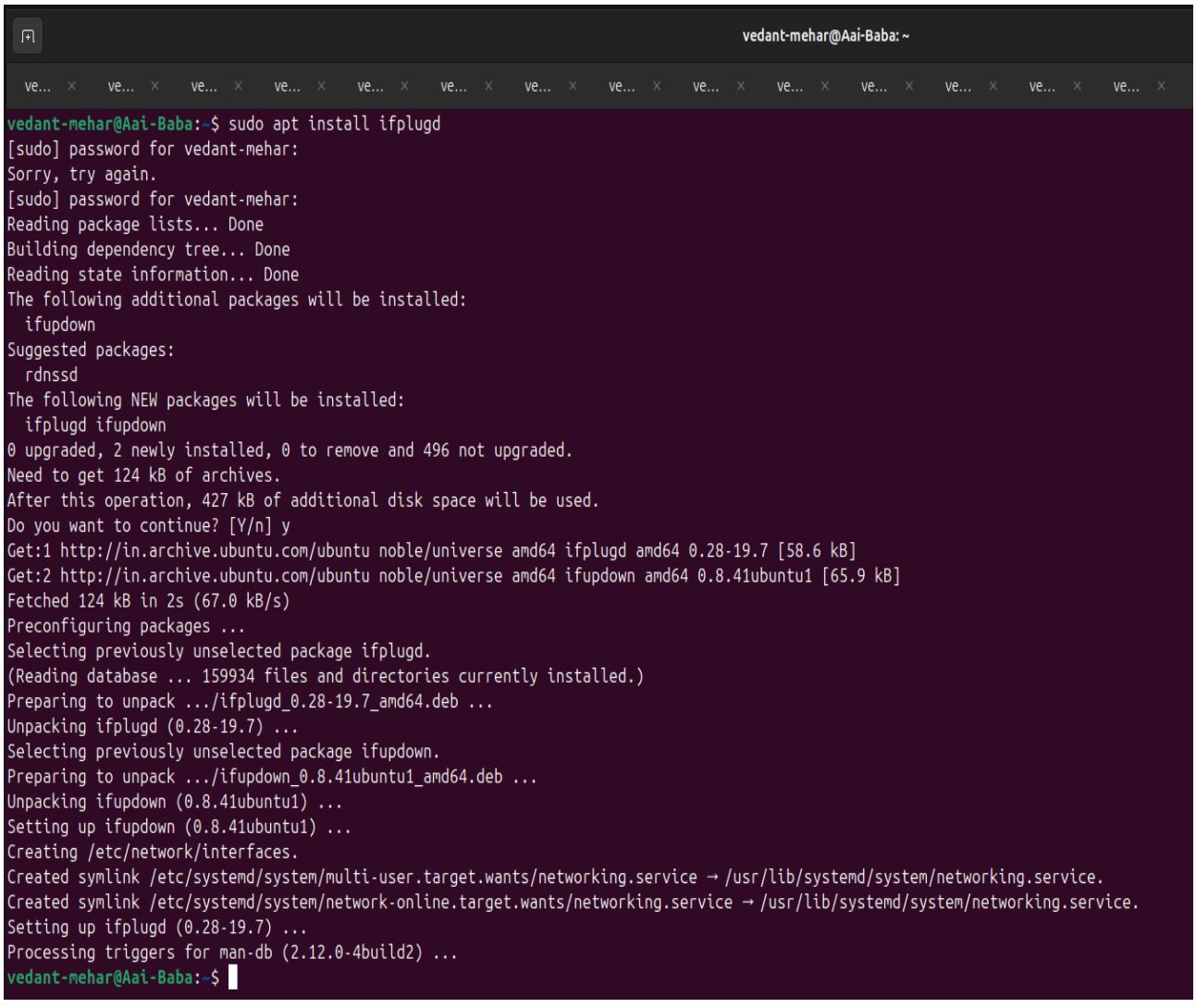
16. mtr

```
                                  My traceroute  [v0.95]
Aai-Baba (10.77.179.140) -> google.com (142.250.134.102)                2026-07-07T13:33:04+0530
Keys:  Help   Display mode   Restart statistics   Order of fields   quit
                                                          Packets               Pings
 Host                                                   Loss%   Snt   Last   Avg  Best  Wrst StDev
 1. _gateway                                             0.0%    18    5.8   5.7   4.1   8.7   1.5
 2. (waiting for reply)
 3. 255.0.0.2                                            5.6%    18   33.5  26.5  16.1  35.0   6.1
 4. 255.0.0.3                                            0.0%    18   30.2  24.9  14.7  31.4   4.6
 5. 192.168.227.67                                       0.0%    18   27.7  25.9  16.2  33.4   4.6
 6. 192.168.92.164                                       0.0%    18   30.7  29.8  17.9  67.2  10.6
    192.168.92.166
 7. (waiting for reply)
 8. (waiting for reply)
 9. 142.250.165.42                                       0.0%    18   37.3  27.6  19.8  37.3   4.0
10. 142.251.225.77                                       0.0%    18   32.3  28.0  23.9  32.3   2.6
11. 192.178.110.104                                      0.0%    18   33.7  26.1  22.1  33.7   2.9
12. 192.178.82.97                                        0.0%    18   35.0  28.4  16.9  36.6   4.8
13. 142.250.208.82                                       0.0%    18   31.1  27.8  20.6  34.3   3.8
14. (waiting for reply)
15. (waiting for reply)
16. (waiting for reply)
17. (waiting for reply)
18. (waiting for reply)
19. (waiting for reply)
20. fx-in-f102.1e100.net                                 0.0%    17   35.7  27.4  14.8  39.7   5.9
```

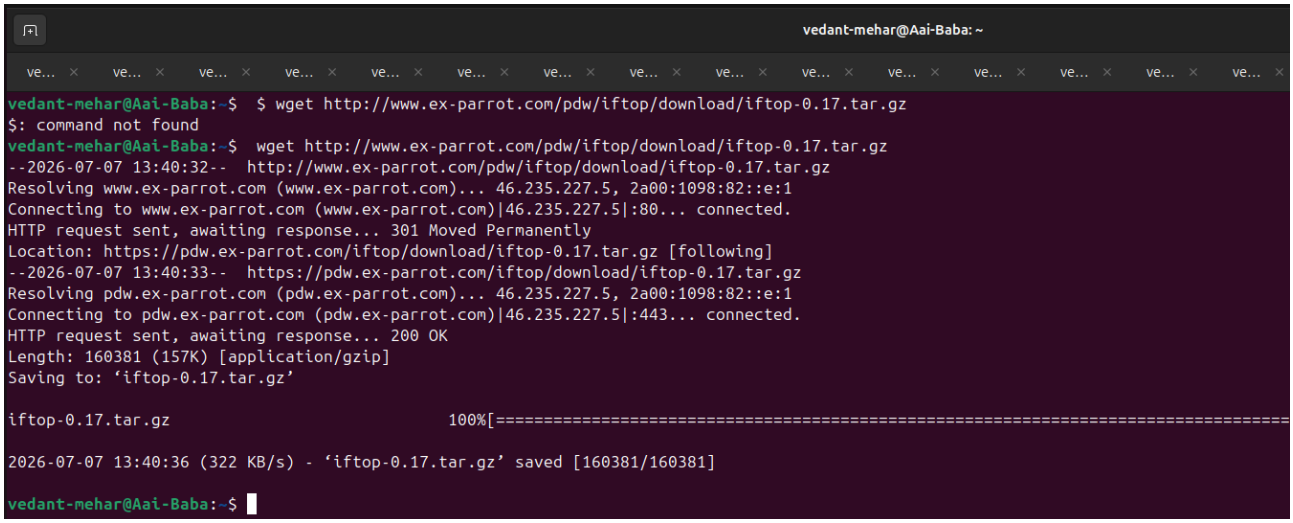
17. whois



18. ifplugstatus

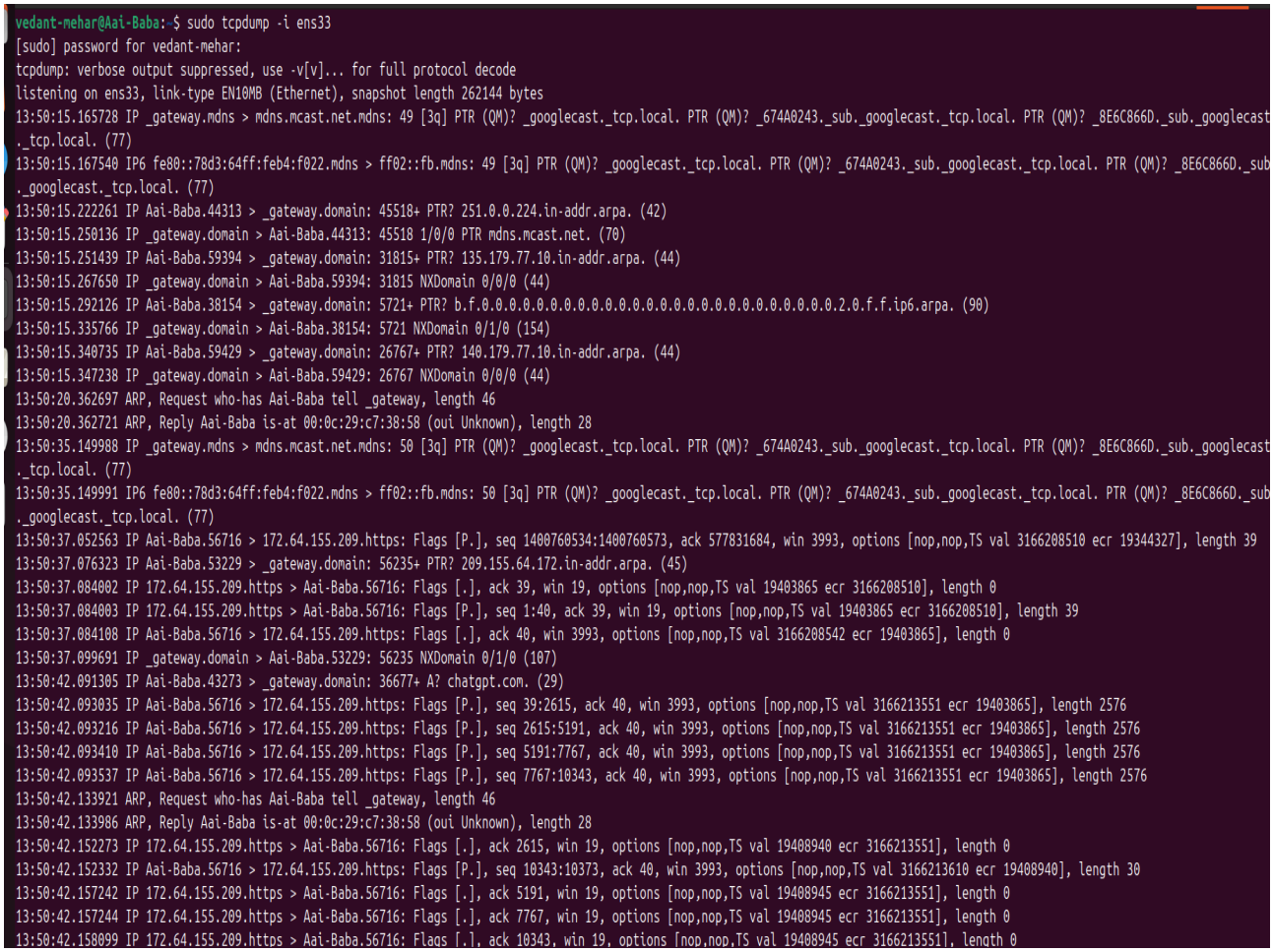


19. iftop


vedant-mehar@Aai-Baba: ~
vedant-mehar@Aai-Baba:~$ $ wget http://www.ex-parrot.com/pdw/iftop/download/iftop-0.17.tar.gz
$: command not found
vedant-mehar@Aai-Baba:~$ wget http://www.ex-parrot.com/pdw/iftop/download/iftop-0.17.tar.gz
--2026-07-07 13:40:32-- http://www.ex-parrot.com/pdw/iftop/download/iftop-0.17.tar.gz
Resolving www.ex-parrot.com (www.ex-parrot.com)... 46.235.227.5, 2a00:1098:82::e:1
Connecting to www.ex-parrot.com (www.ex-parrot.com)|46.235.227.5|:80... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://pdw.ex-parrot.com/iftop/download/iftop-0.17.tar.gz [following]
--2026-07-07 13:40:33-- https://pdw.ex-parrot.com/iftop/download/iftop-0.17.tar.gz
Resolving pdw.ex-parrot.com (pdw.ex-parrot.com)... 46.235.227.5, 2a00:1098:82::e:1
Connecting to pdw.ex-parrot.com (pdw.ex-parrot.com)|46.235.227.5|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 160381 (157K) [application/gzip]
Saving to: ‘iftop-0.17.tar.gz’
iftop-0.17.tar.gz 100%[
2026-07-07 13:40:36 (322 KB/s) - ‘iftop-0.17.tar.gz’ saved [160381/160381]
vedant-mehar@Aai-Baba:~$


checking size of u_int16_t... 2
checking size of u_int32_t... 4
checking for library containing getnameinfo... none required
checking for library containing gethostbyaddr_r... none required
checking how to call gethostbyaddr_r... 8 args, int return
checking gethostbyaddr_r usability... yes
checking where to find pcap.h... no idea
configure: error: can't find pcap.h
You're not going to get very far without libpcap.
vedant-mehar@Aai-Baba:~/iftop-0.17$ make
make: *** No targets specified and no makefile found. Stop.
vedant-mehar@Aai-Baba:~/iftop-0.17$ make install
make: *** No rule to make target 'install'. Stop.
vedant-mehar@Aai-Baba:~/iftop-0.17$ iftop -B
Command 'iftop' not found, but can be installed with:
sudo apt install iftop
vedant-mehar@Aai-Baba:~/iftop-0.17$ sudo apt install iftop
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
iftop
0 upgraded, 1 newly installed, 0 to remove and 496 not upgraded.
Need to get 33.5 kB of archives.
After this operation, 87.0 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 iftop amd64 1.0~pre4-9build2 [33.5 kB]
Fetched 33.5 kB in 1s (32.3 kB/s)
Selecting previously unselected package iftop.
(Reading database ... 160002 files and directories currently installed.)
Preparing to unpack .../iftop_1.0~pre4-9build2_amd64.deb ...
Unpacking iftop (1.0~pre4-9build2) ...
Setting up iftop (1.0~pre4-9build2) ...
Processing triggers for man-db (2.12.0-4build2) ...
vedant-mehar@Aai-Baba:~/iftop-0.17$ iftop -B
interface: ens33
IP address is: 10.77.179.140
MAC address is: 00:0c:29:c7:38:58
pcap_open_live(ens33): ens33: You don't have permission to perform this capture on that device (socket: Operation not permitted)
vedant-mehar@Aai-Baba:~/iftop-0.17$

20. tcpdump



```
vedant-mehar@Aai-Baba:~$ sudo tcpdump -i ens33
[sudo] password for vedant-mehar:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
13:50:15.165728 IP _gateway.mdns > mdns.mcast.net.mdns: 49 [3q] PTR (QM)? _googlecast._tcp.local. PTR (QM)? _674A0243._sub._googlecast._tcp.local. PTR (QM)? _8E6C866D._sub._googlecast
._tcp.local. (77)
13:50:15.167540 IP6 fe80::78d3:64ff:feb4:f022.mdns > ff02::fb.mdns: 49 [3q] PTR (QM)? _googlecast._tcp.local. PTR (QM)? _674A0243._sub._googlecast._tcp.local. PTR (QM)? _8E6C866D._sub
._googlecast._tcp.local. (77)
13:50:15.222261 IP Aai-Baba.44313 > _gateway.domain: 45518+ PTR? 251.0.0.224.in-addr.arpa. (42)
13:50:15.250136 IP _gateway.domain > Aai-Baba.44313: 45518 1/0/0 PTR mdns.mcast.net. (70)
13:50:15.251439 IP Aai-Baba.59394 > _gateway.domain: 31815+ PTR? 135.179.77.10.in-addr.arpa. (44)
13:50:15.267650 IP _gateway.domain > Aai-Baba.59394: 31815 NXDomain 0/0/0 (44)
13:50:15.292126 IP Aai-Baba.38154 > _gateway.domain: 5721+ PTR? b.f.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.2.0.f.f.ip6.arpa. (90)
13:50:15.335766 IP _gateway.domain > Aai-Baba.38154: 5721 NXDomain 0/1/0 (154)
13:50:15.340735 IP Aai-Baba.59429 > _gateway.domain: 26767+ PTR? 140.179.77.10.in-addr.arpa. (44)
13:50:15.347238 IP _gateway.domain > Aai-Baba.59429: 26767 NXDomain 0/0/0 (44)
13:50:20.362697 ARP, Request who-has Aai-Baba tell _gateway, length 46
13:50:20.362721 ARP, Reply Aai-Baba is-at 00:0c:29:c7:38:58 (oui Unknown), length 28
13:50:35.149988 IP _gateway.mdns > mdns.mcast.net.mdns: 50 [3q] PTR (QM)? _googlecast._tcp.local. PTR (QM)? _674A0243._sub._googlecast._tcp.local. PTR (QM)? _8E6C866D._sub._googlecast
._tcp.local. (77)
13:50:35.149991 IP6 fe80::78d3:64ff:feb4:f022.mdns > ff02::fb.mdns: 50 [3q] PTR (QM)? _googlecast._tcp.local. PTR (QM)? _674A0243._sub._googlecast._tcp.local. PTR (QM)? _8E6C866D._sub
._googlecast._tcp.local. (77)
13:50:37.052563 IP Aai-Baba.56716 > 172.64.155.209.https: Flags [P.], seq 1400760534:1400760573, ack 577831684, win 3993, options [nop,nop,TS val 3166208510 ecr 19344327], length 39
13:50:37.076323 IP Aai-Baba.53229 > _gateway.domain: 56235+ PTR? 209.155.64.172.in-addr.arpa. (45)
13:50:37.084002 IP 172.64.155.209.https > Aai-Baba.56716: Flags [.], ack 39, win 19, options [nop,nop,TS val 19403865 ecr 3166208510], length 0
13:50:37.084003 IP 172.64.155.209.https > Aai-Baba.56716: Flags [P.], seq 1:40, ack 39, win 19, options [nop,nop,TS val 19403865 ecr 3166208510], length 39
13:50:37.084108 IP Aai-Baba.56716 > 172.64.155.209.https: Flags [.], ack 40, win 3993, options [nop,nop,TS val 3166208542 ecr 19403865], length 0
13:50:37.099691 IP _gateway.domain > Aai-Baba.53229: 56235 NXDomain 0/1/0 (107)
13:50:42.091305 IP Aai-Baba.43273 > _gateway.domain: 36677+ A? chatgpt.com. (29)
13:50:42.093035 IP Aai-Baba.56716 > 172.64.155.209.https: Flags [P.], seq 39:2615, ack 40, win 3993, options [nop,nop,TS val 3166213551 ecr 19403865], length 2576
13:50:42.093216 IP Aai-Baba.56716 > 172.64.155.209.https: Flags [P.], seq 2615:5191, ack 40, win 3993, options [nop,nop,TS val 3166213551 ecr 19403865], length 2576
13:50:42.093410 IP Aai-Baba.56716 > 172.64.155.209.https: Flags [P.], seq 5191:7767, ack 40, win 3993, options [nop,nop,TS val 3166213551 ecr 19403865], length 2576
13:50:42.093537 IP Aai-Baba.56716 > 172.64.155.209.https: Flags [P.], seq 7767:10343, ack 40, win 3993, options [nop,nop,TS val 3166213551 ecr 19403865], length 2576
13:50:42.133921 ARP, Request who-has Aai-Baba tell _gateway, length 46
13:50:42.133986 ARP, Reply Aai-Baba is-at 00:0c:29:c7:38:58 (oui Unknown), length 28
13:50:42.152273 IP 172.64.155.209.https > Aai-Baba.56716: Flags [.], ack 2615, win 19, options [nop,nop,TS val 19408940 ecr 3166213551], length 0
13:50:42.152332 IP Aai-Baba.56716 > 172.64.155.209.https: Flags [P.], seq 10343:10373, ack 40, win 3993, options [nop,nop,TS val 3166213610 ecr 19408940], length 30
13:50:42.157242 IP 172.64.155.209.https > Aai-Baba.56716: Flags [.], ack 5191, win 19, options [nop,nop,TS val 19408945 ecr 3166213551], length 0
13:50:42.157244 IP 172.64.155.209.https > Aai-Baba.56716: Flags [.], ack 7767, win 19, options [nop,nop,TS val 19408945 ecr 3166213551], length 0
13:50:42.158099 IP 172.64.155.209.https > Aai-Baba.56716: Flags [.], ack 10343, win 19, options [nop,nop,TS val 19408945 ecr 3166213551], length 0
```